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Cyber Criminals Hack the Hops



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Cyber Criminals
Hack the Hops



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01 Introduction

On a global scale, cyber criminals have used ransomware to extract a king's ransom from successful enterprises. It appears that they will attack any industry, worts[1] and all, from breweries to meat processors to the entire food supply chain according to a warning from the FBI in the USA.

A surprising new target for cyber attack was revealed when cyber criminals hit Lion brewery with a ransomware attack in 2021, just as Australia was easing COVID-19 restrictions and re-opening pubs. All beer processing systems were immediately taken offline and Lion was forced to implement manual processes to handle orders and deliveries.[2] Earlier that same year, brewing giant Molson Coors filed an SEC Form 8-K describing a cybersecurity incident that disrupted beer production. They hired an IT forensic firm to investigate the breach. Shipments were delayed while systems were brought back online. The Campari Group Italian beverage vendor was hit and production at Arizona Beverages was also knocked offline because of ransomware attacks.[3] Barcelona's Damm beer maker stopped production due to a cyber attack at its main brewery.[4]

Ransomware halted meat processing at a USA plant for a day and temporarily shut down production in Australia and Canada before the ransom was finally paid.[5] A global leader in the manufacture and distribution of agricultural machinery was hit with a ransomware attack in May 2022. Data was stolen and the full extent of the damage has not been determined.[6]

1 All beer begins with wort. Brewers steep malted barley in water, add hops, then boil the mixture which results in wort. Yeast is added to the wort and the liquid then ferments to produce beer.

2 Lee Mathews, Ransomware Attack Shuts Down Brewing Giant Just As Coronavirus Restrictions Ease, Forbes (2020), https://www.forbes.com/sites/leemathews/2020/06/12/ransomware-attack-shuts-down-brewing-giant-just-as-coronavirus-restrictions-ease/?sh=332c09cc6a69 (accessed May 20, 2022)

3 Natalie Gagliordi, "Molson Coors discloses cyberattack disrupting its brewery operations" , ZDNet (2021), https://www.zdnet.com/article/molson-coors-discloses-cyberattack-disrupting-its-brewery-operations/ (access May 20, 2022)

4 Cyber attack turns off the taps at Barcelona' s Damm brewery (2021) https://www.reuters.com/world/europe/cyber-attack-turns-off-taps-barcelonas-damm-brewery-2021-11-12/ (accessed May 20, 2022)

5 Meat giant JBS pays $11m in ransom to resolve cyber-attack, BBC (2021) https://www.bbc.com/news/business-57423008 (access May 23, 2022)

6 Greg Peterson, "News Release: AGCO Provides Update on Recovery from Ransomware Cyber Attack" , AGCO (2022) https://news.agcocorp.com/news/agco-provides-update-on-recovery-from-ransomware-cyber-attack (accessed May 23, 2022)

02

Cascading Ransomware Attacks Food Supply Chain

What does beer production have in common with meat processing and agricultural machinery? Disrupting the food supply chain can start in the field or at the farm. Grain is used to feed animals and humans, while barley is used to make beer. Cattle ranches and dairy farms give us meat and milk. Holding our raw food supply for ransom would have a cascading effect, from empty shelves at the grocery store to plummeting financial markets where commodities are traded.

In March 2022, a multi-state grain company suffered a Lockbit 2.0 ransomware attack. This company provides grain along with seed, fertilizer, and logistics services that are critical during the spring planting season. In February 2022, a company providing feed milling and other agricultural services reported two instances of unauthorized access to some systems. Investigators believe bad actors were attempting to plant ransomware. Fortunately, these attempts were detected before files were encrypted. Between 15 September and 6 October 2021, six grain cooperatives suffered ransomware attacks from a variety of ransomware, including Conti, BlackMatter, Suncrypt, Sodinokibi, and BlackByte. In July 2021, a business management software company discovered the HelloKitty (also known as 'Five Hands') ransomware in their system after the threat actor demanded $30 million. This led to ransomware infections downstream, hitting several of their agricultural cooperative customers.

03 5 Steps from Farm to Plate

The five basic steps for processing food products include:

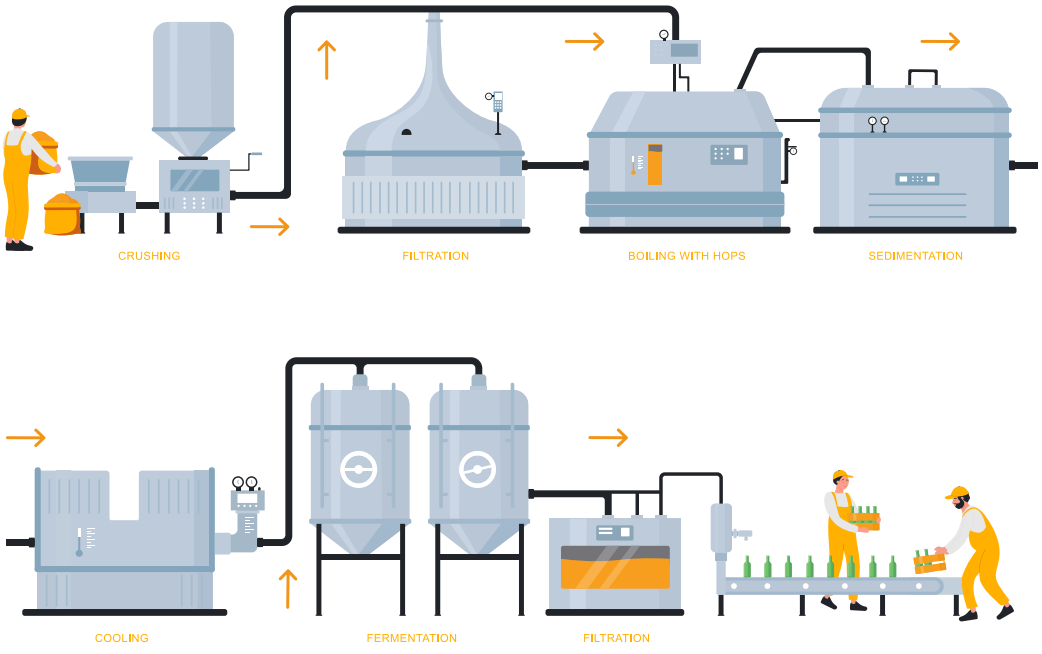
1. Planting, growing, and harvesting vegetables, grains, nuts, and other plant-based foods. Farm equipment is becoming increasingly autonomous. Tractors and pickers rely on GPS to navigate fields. Sensors monitor the feeding the and watering of plants, along with weather conditions and environmental factors.
2. Raising cattle, pigs, chickens and other animals. Technology is changing the way animals are fed, watered, treated, and slaughtered.
3. Transporting raw foods to the processing plant. Autonomous trucks are drive crops from the farm to the processing plant.
4. Processing raw food into edible products. Every type of food product has its own unique process. In a potato processing plant, potatoes travel on conveyor belts through a series of machines where they are peeled, sliced, and diced before being cooked in big vats. After that, the cooked potatoes can be processed into flakes, baked, or fried before being frozen. Processing fermented products involves chemical processing that can be more dangerous than processing other food items. Alcoholic fermentation is the complex system of interactions between the yeast and the wort. Bioreactors are used for more than 80% of fermented products including beer. If not operated according to safety procedures, fermentation tanks and bioreactors can explode.
5. Once processed, food products are packed onto delivery trucks and shipped to grocery stores, restaurants, bars, hospitals, and other places that sell or serve food.

All of these processes involve a wide range of technologies from legacy devices still running Windows XP to advanced robotics driving autonomous tractors. More and more of these technologies are interconnected through the internet. Control commands are sent via network protocols. Because our food supply, including beer, is so valuable, bad actors put a tremendous amount of time into plotting new ways to inject ransomware and score a big payday.



04 Smart Beer

The beer industry uses many of the same systems as food processing in addition to special equipment for fermentation. Barley is harvested at the farm and trucks transport it to the brewery. Brewers steep malted barley in water. They add hops and boil the mixture, resulting in wort. Bioreactors grow yeast. Fermentation is the process by which yeast converts the glucose in the wort to ethyl alcohol and carbon dioxide gas. This gives beer its alcohol content and carbonation.



The following diagram shows where the automated controls could be compromised.[7] A bad actor could attack the mashing process and increase the temperature to cause an explosion.

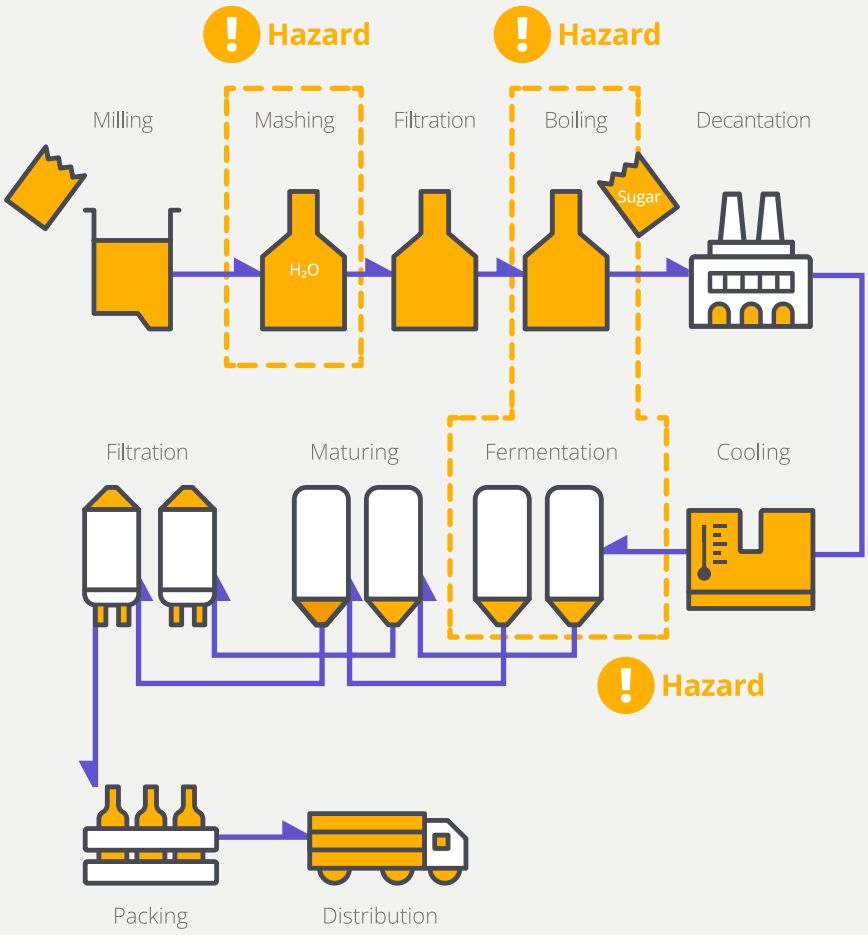


Fig. 1. The general process of beer production and the areas of potentials hazards

7 Valdrin M. Beluli, "Smart beer production as a possibility for cyber-attack within the industrial process in automatic control", Procedia Computer Science 158 (2109) 206-213, www.ScienceDirect.com (accessed May 20, 2022)

05 Tanks

Traditionally, beer fermentation was done in open fermentation and lager tanks. These have been replaced by large enclosed tanks that have a pipe to release CO2. If the CO2 is not released properly, the tank can explode. Modern CO2 release valves are electronic. A bad actor could cause an explosion by taking control of the CO2 release valve at the fermentation tank.

The type of sensor used to monitor the tank is important. In one study, traditional IT antivirus software was monitoring the tank control system. The antivirus reported "all clear" when in fact, biochemical reactions were building up pressure inside the tank because a virus had attacked the valve of the electronic CO2 release. The tank exploded.

Traditional antivirus software has not been designed to interact with cyber-physical systems. OT zero trust was built with operational technology in mind. It can understand the protocols that tank sensors use to communicate. For example, research indicates that a laser sensor can detect metal expansion caused by high vapor pressure or carbon dioxide that indicate an explosion may be brewing. The laser is attached on the side of the tank. If the laser beam is broken, then it sends an immediate alert that the tank is expanding beyond normal limits. OT zero trust can read these alerts and broadcast them across the plant or trigger a safety system in response.

In addition to tank pressure, temperature must also be carefully monitored. When temperatures start rising, sensors send alerts and OT zero trust can respond. In the past, machines have relied on mechanical systems that automatically shut down the machine when a safety threshold is breached. These mechanical security systems may still provide a second layer of defense, but OT zero trust allows you to see what's going on across the entire plant. This makes it possible to take other safety actions, such as quarantining infected systems. It also provides plant-wide data useful for analyzing and modeling efficiencies as well as threat intelligence to stop attacks before they get started.

06 Bioreactors

Brewers use bioreactors to grow yeast. In addition to brewing beer, bioreactors are used in the production of pharmaceuticals and vaccines, as well as for turning corn into ethanol for fuel. A cyber attack that increases the temperature of a bioreactor beyond safety limits can cause an explosion. All the attacker needs to know are the details of the communication protocol used by the bioreactor and the system dynamics.

Bioreactor temperature sensors keep track and report the temperature, but they cannot determine whether malware has been injected. Operators follow Standard Operating Procedures (SOP) that may require applying safe cell chemistry, a supervisory disaster-control system, or additional sensor technology for stand-alone monitoring to guard against explosions.

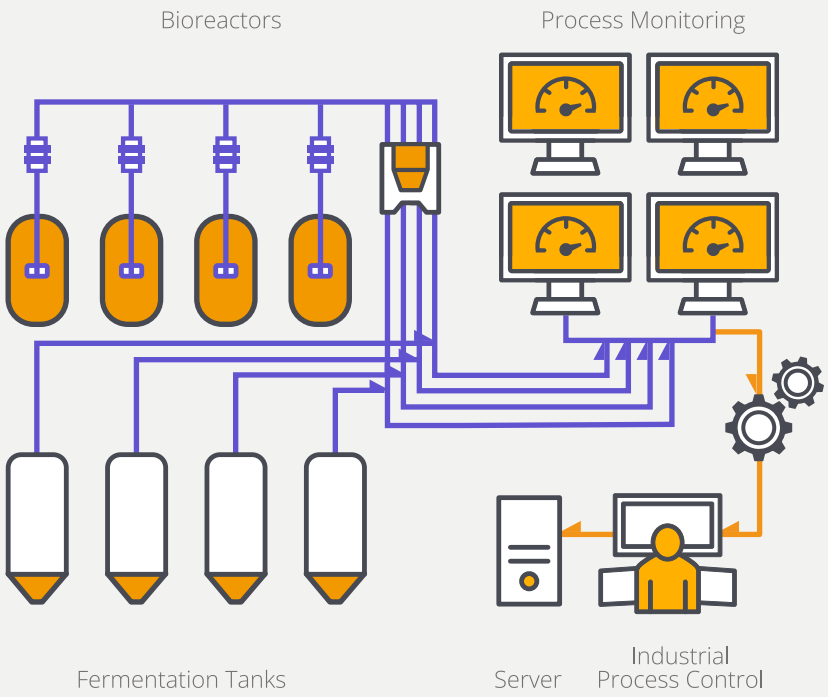


Fig. 2. Automatic control of the brewing process by the operators

07 OT Zero Trust

In this potentially volatile environment, you need a cyber security tool built to interact with sensors and operational technology. OT zero trust was built to understand the communications protocols of machines. Endpoint protection can understand sensor messages and send alerts or take protective action. OT Firewalls can stop malicious traffic by comparing network traffic to trust lists.[8]

To prevent ransomware, the FBI recommends:[9]

1. Segmenting networks
2. Regularly updating malware protection on endpoints
3. Installing updates and patching operating systems, software, and firmware as soon as patches are released
4. Using strong access controls and multi-factor authentication where possible
5. Developing a recovery plan that includes manual operation, if necessary
6. Conducting security audits
7. Conducting cybersecurity awareness training
8. Using secure networks such as VPNs

OT zero trust was built for operational technology. It was designed to meet FBI recommendations for segmenting networks, protecting endpoints, patching systems, and conducting security audits and inspections. Additionally, security logs from OT zero trust security devices support developing a recovery plan and cybersecurity training.

OT zero trust takes cybersecurity further by adding virtual patching to protect against zero-day attacks and threat intelligence to outsmart attackers by predicting where they may attack next. One of the key benefits of OT zero trust is extending the life of heirloom systems – these systems that are no longer being supported by vendors are too valuable to abandon.

8 Valdrin M. Beluli, "Smart beer production as a possibility for cyber-attack within the industrial process in automatic control", Procedia Computer Science 158 (2109) 206-213, www.ScienceDirect.com (accessed May 20, 2022)

9 Ransomware Attacks on Agricultural Cooperatives Potentially Timed to Critical Seasons, Private Industry Notification Federal Bureau of Investigation Cyber Division (FBI:2022) https://www.ic3.gov/Media/News/2022/220420-2.pdf (access May 23, 2022)

08 Securing Air-Gapped and Legacy Devices

Manufacturing ice cream, mayonnaise, and other food products requires safeguarding manufacturing plants in many countries around the world. While processes in each plant may be unique, the first step in cybersecurity is identifying all the cyber assets used in the manufacturing process.[10] Not all factories are equal.[11] Understanding similarities and differences informs the deployment of security patches and monitoring of network traffic so that each device only receives trusted command controls and cannot be infected by ransomware. For example, some devices are still running on legacy operating systems such as Windows XP, and others cannot connect to the internet.

The good news is that it is still possible to protect these devices. The OT zero trust asset inspector looks like a USB memory stick, but it is a powerful malware scanner that also takes inventory of all the software, firmware, and computer information onboard any device without installing any special software or slowing down production. The inventory is automatically uploaded to the OT defense console where it is aggregated with data from other devices to paint a picture of the entire factory. Legacy and air-gapped devices can be scanned periodically or on-demand. For those assets that can access the internet, advanced endpoint protection uses allowlists to shield them from malicious messages such as ransomware attacks.

10 Unilever's Bobby Ford Discusses How to Secure Your Digital Business, Wall Street Journal (2020) https://www.getabstract.com/es/resumen/unilevers-bobby-ford-discusses-how-to-secure-your-digital-business/41711

11 *Unilever Focuses on Factory Cybersecurity as Pandemic Sparks Run on Consumer Staples*, SCCEU (2020) https://scceu.org/unilever-focuses-on-factory-cybersecurity-as-pandemic-sparks-run-on-consumer-staples/ (accessed May 23, 2022)

09 Future Trends

Hop is a volatile ingredient that must be added to the brewing process within 12 hours of being harvested. IoT-enabled sensors can help ensure that hop is harvested in peak conditions.

Breweries are adopting IoT, AI, machine learning, big data, edge computing, virtual reality (VR) and augmented reality (AR) to create more automated production processes across different departments, which include product design, packaging, manufacturing, QA and logistics departments.

Digital twins are being created for brewery products, plants, and production lines to aid in simulating and optimizing the entire workflow.

Smart Beer Membrane Filter (BMF) gathers data about the best time to clean the filter and remove the particles that clog it up. This makes maintenance more cost effective. IoT sensors on packaging machines record changes in temperature or vibrations. It enables predictive maintenance and reduces downtime of the production line.

Every time a new IoT device is introduced into the plant, another security challenge is also introduced. Inspect every new IoT device before it is deployed using the OT zero trust portable security devices to wipe any malware and automatically take inventory of all software, computer information, and firmware on the IoT device.



10 Appendix – Alcohol Regulatory Guidelines

OT zero trust includes a risk assessment toolkit to aid in regulatory compliance efforts.

Regulation (EU) 1169/2011 (EU 2011) requires that the actual alcoholic strength by volume be explicitly written on the label for alcoholic beverages that contain more than 1.2% by volume of alcohol.

EU Directive 2010/13/EU sets criteria for commercial communication (television advertising, teleshopping, sponsorship, and product placement) of alcohol and other commodities.

Federal Alcohol Administration Act (27 U.S.C., Chapter 8, section 201-212) was enacted in 1935 to further protect the revenue derived from distilled spirits, wine, and malt beverages, to regulate interstate and foreign commerce.

Malt beverage is defined as a beverage made by the alcoholic fermentation of specific materials [27 U.S.C. 211(a)(7)]. Wine is defined in FAA Act as containing not less than 7 and not more than 24 percent alcohol by volume [27 U.S.C. 211(a)(6)].



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